

CELL AND MOLECULAR BIOLOGY LABORATORY
From Gene Mutation to Disease

Doreen Marie G. Puyat	Mrs. Lilibeth Bucol & Mr. Abner Bucol
BIO 300 - Cell and Molecular Biology	Disease: Spinocerebellar ataxia type 3 Gene: ATXN3

PART I. DISEASE BACKGROUND

Spinocerebellar ataxia type 3 (SCA3), also called Machado-Joseph disease, is a progressive autosomal dominant neurodegenerative disorder and the most common inherited ataxia worldwide. It is caused by an expansion of a CAG trinucleotide repeat in the ATXN3 gene on chromosome 14, which translates into an abnormally long polyglutamine tract in the ataxin-3 protein. Normal alleles have about 12–44 CAG repeats, while pathogenic alleles expand to roughly 52–86 repeats, with longer repeats linked to earlier and more severe disease. The disorder primarily damages the cerebellum, brainstem, and spinal cord, leading to progressive loss of motor coordination. Clinically, patients present with ataxia (unsteady gait and limb movement), dysarthria (slurred speech), and abnormal eye movements such as nystagmus. Because SCA3 is autosomal dominant, a single expanded allele is enough to cause disease, and the condition shows genetic anticipation, where repeat length can increase across generations. Onset is typically in adulthood, with symptoms worsening progressively over time.

PART II. RESEARCH DISEASE

A. Disease

What is the disease or phenotype?	Spinocerebellar ataxia type 3 (SCA3)
What are its major clinical characteristics?	Progressive gait/limb incoordination, dysarthria, abnormal eye movements (nystagmus, ophthalmoplegia), and in some subtypes pyramidal/extrapyramidal signs or parkinsonism. It's a late-onset autosomal dominant neurodegenerative disorder.
Which cells, tissues, or organs are mainly affected?	Cerebellum, brainstem (dentate nucleus, pontine nuclei), and spinal cord, with the ataxin-3 protein forming aggregates in these regions.

What is its genetic basis?	It is caused by an expanded XAG triplet repeat in the ATXN3 gene, encoding a polyglutamine (polyQ) tract.
What is its inheritance pattern, if applicable?	It is autosomal dominant with anticipation (repeat lengths tends to increase across generations, especially paternal transmission).

B. Gene and Normal Protein

What is the official gene symbol?	ATXN3
On which human chromosome is the gene located?	14q32.12 (also written 14q24.3–q32.2 in older sources)
What does the gene normally encode?	Ataxin-3, a deubiquitinating enzyme (DUB) that trims long polyubiquitin chains to maintain protein homeostasis.
What is the normal biological function of the protein?	An N-terminal Josephin domain responsible for the ubiquitin hydrolase catalytic activity, and a flexible C-terminal region containing ubiquitin-interacting motifs (UIMs) plus the polyQ tract encoded by the CAG repeat.
Where in the cell is the protein normally found?	It operates in the ubiquitin-proteasome system and connects protein quality control with transcriptional regulation, autophagy, and mTORC1 signaling — e.g., stabilizing BECN1 and inactivating RHEB.
In what biological pathway or cellular process does it participate?	Nucleus, nucleoplasm, cytosol, and lysosomal membrane.

C. Documented Mutation

Gene	ATXN3
Reference transcript	NM_004993.6
Exact variant notation	NM_004993.6:c.892CAG[8_36]
Nucleotide change	Expansion of the CAG trinucleotide repeat at c.892 from a normal range of 13–36 repeats to a pathogenic range of 68–79 repeats
Predicted protein change	Elongation of the polyglutamine (polyQ) tract in ataxin-3
Mutation type	Trinucleotide repeat expansion (in-frame)
ClinVar accession when available	RCV000003729
Clinical interpretation	Pathogenic — causes Machado-Joseph disease (spinocerebellar ataxia type 3)
Scientific references supporting the association	Kawaguchi, Y. et al. (1994). CAG expansions in a novel gene for Machado-Joseph disease at chromosome 14q32.1. <i>Nature Genetics</i> , 8(3), 221–228.

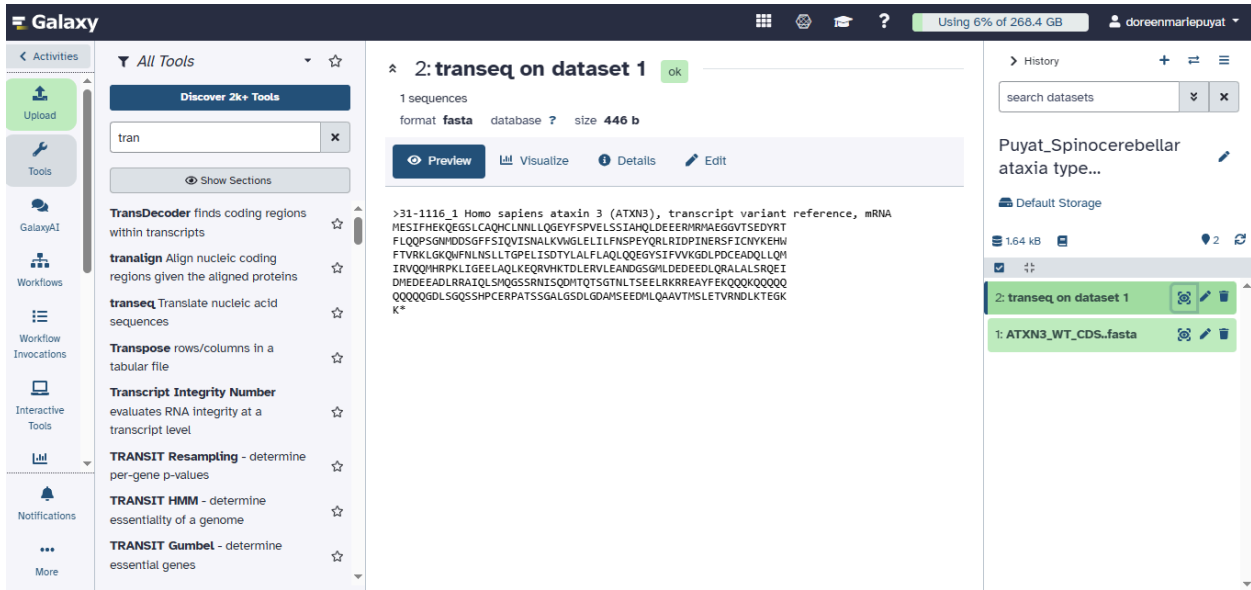


Figure 1. Transeq translation output of the wild-type ATXN3 CDS in Galaxy, showing the predicted 361-aa protein sequence with intact start/stop codons and a normal (13-residue) polyglutamine tract.

Part V. ESTABLISH THE WILD-TYPE CONTROL

CDS length	1086 bp
Predicted protein length	361 amino acids
Start codon	ATG
Stop codon	TAA
Reading frame	+1 (Frame 1)
First 10 aminoacids	MESIFHEKQE
Last 10 amino acids	RNDLKTEGKK
Comparison between predicted protein and the accepted reference protein	The predicted protein sequence is identical to the accepted reference protein sequence (NP_004984.2).

Part VI. FORMULATE A MUTATION HYPOTHESIS

Mutation and exact nucleotide change	1-nucleotide deletion from the WT ATXN3 CDS.
Number of nucleotide(s) affected	1 nucleotide deleted
Predicted mutation type	Frameshift mutation
Predicted effect on the reading frame	The reading frame will shift
Predicted effect on protein length	It is expected to be altered and likely shortened
Predicted effect on protein function	It is expected to change many downstream amino acids and may produce an abnormal/nonfunctional protein

Part VII. CREATE THE MUTANT SEQUENCE

Original nucleotide position (s)	c.426
Original sequence	GACAGTGGTTTACAGTTAGAAA
Mutant sequence	GACAGTGGTTTAAGTTAGAAA
Number of bases inserted/deleted/substituted	1 nucleotide deleted
Mutation type	Frameshift

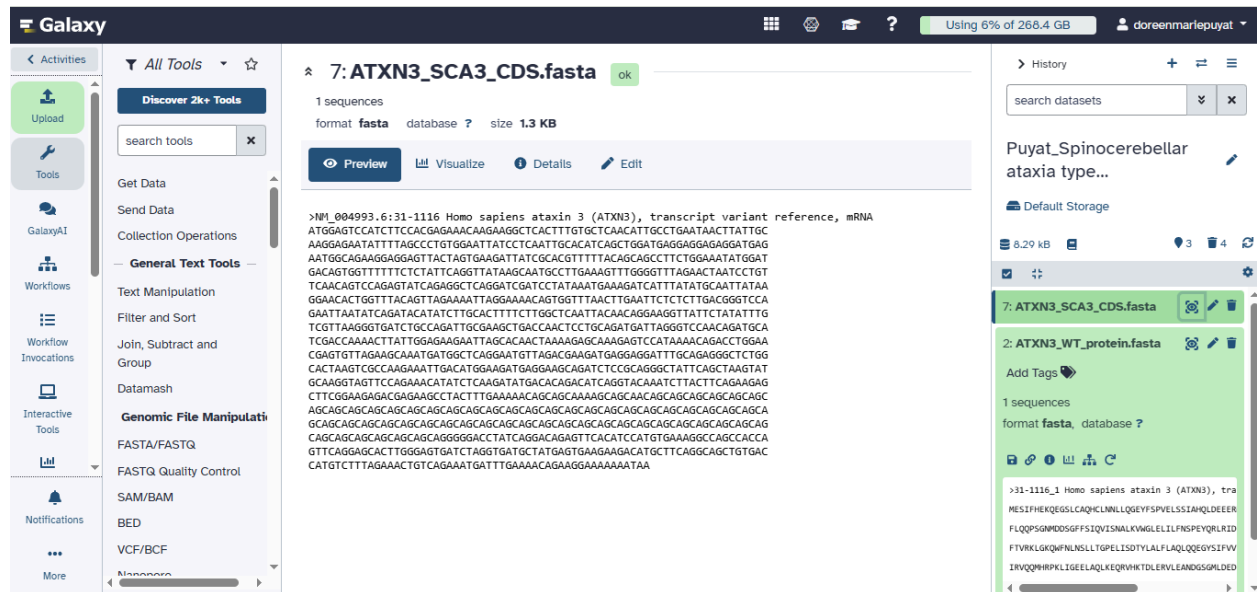


Figure 2. ATXN3_SCA3_CDS.fasta containing the modified ATXN3 CDS sequence with 54 additional CAG repeats uploaded to Galaxy for analysis.

Part VIII. TRANSLATE THE MUTANT SEQUENCE

Mutant CDS length	1248 bp
Mutant protein length	415 aa
Reading frame	+1 / Frame 1
Location of the first amino- acid difference	Approximately amino acid 62
Presence or absence of a premature stop codon	Present
Approximate number of amino acids affected	Approximately 300 amino acids downstream of the deletion

Part IX. COMPARE WT AND MUTANT PROTEIN

At what amino-acid position do the sequences first differ?	Position 301
Is only one amino acid affected?	No
Are multiple downstream amino acids changed?	No
Was an amino acid deleted or inserted?	Inserted
Was a premature stop codon produced?	No

Did the reading frame change?	No
Did the protein length change?	Yes, 361 aa → 415 aa
Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?	Trinucleotide repeat expansion

Figure 3. Combined sequence datasets generated by concatenating Dataset 2 and Dataset 9 in Galaxy for downstream analysis.

Part X. EXPLAIN THE MOLECULAR CONSEQUENCE

The documented mutation in SCA3 is a CAG trinucleotide repeat expansion in the coding region of the ATXN3 gene. In this experiment, the wild-type ATXN3 CDS was 1,086 bp long, while the modeled mutant sequence contained 54 additional CAG repeats, increasing the CDS length to 1,248 bp. Because each CAG consists of three nucleotides, the insertion was divisible by three and therefore did not change the reading frame. At the protein level, CAG encodes glutamine (Q). Therefore, the additional CAG repeats produced an expanded polyglutamine tract in the mutant ATXN3 protein. The WT protein was 361 amino acids long, whereas the mutant protein was 415 amino acids long, representing 54 additional glutamine residues. The sequence alignment showed that the main difference between the proteins was the longer Q-rich region, while the downstream C-terminal sequence remained aligned and the normal terminal stop codon was retained.

The expanded polyglutamine tract can affect the properties of ataxin-3. Published evidence indicates that expanded polyglutamine ATXN3 is prone to misfolding, aggregation, and abnormal localization in

neurons. These changes can interfere with cellular protein homeostasis and other cellular processes. Because neurons are particularly vulnerable to these effects, accumulation and dysfunction of expanded ATXN3 are associated with neuronal degeneration and the neurological features of SCA3.

Part XI. SECOND EXPERIMENT: CREATE YOUR OWN MUTATION

Artificial mutation	1 bp deletion
WT CDS length	1,086 bp
Artificial mutant CDS length	1,085 bp
Reading frame	Changed/frameshift
Amino acids affected	Multiple downstream amino acids
Premature stop codon	Yes
Protein sequence	Altered downstream sequence followed by premature codons
Mutation type	Frameshift mutation

The screenshot displays the Galaxy web interface. The main panel shows a FASTA file named "15: ATXN3_artificial_1bp_deletion_CDS.fasta.txt" with 1 sequence, format "fasta", database "?", and size "1.3 KB". The sequence is displayed in a monospace font, showing a 1 bp deletion in the ATXN3 CDS sequence. The right sidebar shows a history of datasets, including "16: transeq on dataset 15", "15: ATXN3_artificial_1bp_deletion_CDS.fasta.txt", "14: MAFFT on dataset 13", "13: Concatenate multiple datasets or collections on datasets 2 and 9", "9: ATXN3_SCA3_MUTANT_protein.fasta", and "8: ATXN3_SCA3_CDS.fasta".

Figure 4. Screenshot showing ATXN3 artificial mutant, the deletion of one nucleotide from a copy of the WT CDS sequence.

For the second experiment, one cytosine (C) was deleted from the ATXN3 coding sequence at CDS position 396 (c.426 in NM_004993.6 numbering), giving the variant NM_004993.6:c.426delC. This is a

1-nucleotide deletion, one of the required options for the artificial mutation experiment. Since 1 is not divisible by 3, this deletion is expected to shift the reading frame from that point onward. All downstream codons will be regrouped, changing many amino acids after the deletion site and likely introducing a premature stop codon, resulting in a shortened, non-functional protein.

This mutant sequence was then translated in Galaxy to test whether the predicted frameshift and premature stop actually occurred.

Part XII. COMPARE THE DOCUMENTED AND ARTIFICIAL MUTATIONS

	WT Sequence	Documented mutation	Student-created mutation
CDS length	1,086 bp	1,248 bp	1,085 bp
Protein length	361 aa	415 aa	Altered/shortened
Mutation type	No mutation	CAG trinucleotide repeat expansion	1-nucleotide deletion (c.426delC)
Whether the reading frame changed	No	No	Yes
Whether a premature stop codon appeared	No	No	Yes
Amino acids affected	None	54 additional glutamine (Q) residues	Many downstream amino acids are changed
Expected functional consequence	Normal ATXN3 protein sequence and expected normal function	Expanded polyglutamine tract may alter ATXN3 protein folding, aggregation, and cellular behavior associated with SCA3	Likely severe disruption of the protein because the reading frame changes, producing altered downstream amino acids and premature termination

The WT ATXN3 protein remained unchanged, while the documented mutation added 54 glutamine residues without shifting the reading frame. In contrast, the artificial 1-nucleotide deletion caused a frameshift, changing downstream amino acids and producing a premature stop codon.

Part XIII. INTERPRETATION QUESTIONS

32. Why does the exact location of a mutation matter?

- The location matters because it determines which codon, amino acid, or functional region of the protein is affected. A mutation in an important region may have a bigger effect.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- Deleting three nucleotides can remove one amino acid without changing the reading frame. Deleting one or two nucleotides shifts the reading frame and changes many downstream amino acids.

34. Does every mutation change the amino-acid sequence? Explain.

- No. Some mutations are synonymous, meaning the codon changes but still codes for the same amino acid.

35. Does every amino-acid substitution destroy protein function? Explain.

- No. Some amino-acid substitutions have little or no effect, especially if the new amino acid has similar properties.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- Because deleting one nucleotide changes how all the following nucleotides are grouped into codons, altering many downstream amino acids.

37. Why might a premature stop codon produce a nonfunctional protein?

- It can stop translation too early, producing a shortened protein that may be missing important parts needed for its function.

38. Could a mutation affect protein function without greatly changing protein length?

- Yes. For example, a single amino-acid change can alter an important site or the protein's shape without greatly changing its length.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes. A mutation can affect gene regulation, such as changing how much protein is produced or when the gene is expressed.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- On my analysis showed that the SCA3 mutation adds 54 CAG repeats, producing 54 additional glutamine residues and increasing the ATXN3 protein from 361 to 415 amino acids. This supports the mechanism of an expanded polyglutamine region.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- My computational results directly show the DNA sequence change, CDS length, protein length, reading frame, and amino-acid changes. And the actual effects on protein folding, aggregation, cell function, and disease require evidence from published experimental studies.

Limitation

This study was based mainly on in-silico sequence analysis, so the results show predicted effects of the mutations at the DNA and protein-sequence levels. The analysis could identify changes in nucleotide sequence, reading frame, amino-acid sequence, and protein length, but it could not directly confirm changes in protein folding, aggregation, cellular function, or disease progression. Experimental laboratory studies would be needed to verify these biological effects.

Conclusion

The analysis demonstrated how different types of mutations can produce different effects on a protein. The documented ATXN3 CAG repeat expansion increased the coding sequence by 162 bp and added 54 glutamine residues without changing the reading frame, while the artificial c.426delC deletion shifted the reading frame and altered the downstream protein sequence. Overall, the experiment showed how the type and size of a mutation can directly influence the resulting protein sequence and its potential functional consequences.

Part XIV. GALAXY AND GITHUB DOCUMENTATION

GitHub repository:

https://github.com/Doreenmariepuyat/Spinocerebellar_ataxia_type_3_ATXN3_Mutation_Lab

Galaxy link:

<https://usegalaxy.org/u/doreenmariepuyat/h/puyat-spinocerebellar-ataxia-type-3-atxn3-mutation-lab>

References

Journal articles

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Database records

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